

Loops in C

Definitions, Concepts & a 40-Program Practice Set
(10 Basic Programs + 30 Pattern Programs)

1. Program List Overview

The next sections contain **40 C programs: 10 basic logic-building programs** followed by **30 pattern-printing programs**. Every program shows the exact console output it produces.

Basic Programs (1-10)

1. Sum of First N Natural Numbers
2. Factorial of a Number
3. Multiplication Table
4. Sum of Digits of a Number
5. Reverse a Number
6. Check Prime Number
7. Fibonacci Series
8. Check Palindrome Number
9. GCD of Two Numbers
10. Print Even and Odd Numbers in a Range

Pattern Programs (1-30)

1. Right Triangle (Solid Star Pattern)
2. Right-Aligned Triangle
3. Inverted Right Triangle
4. Inverted Right-Aligned Triangle
5. Pyramid Pattern
6. Inverted Pyramid Pattern
7. Diamond Pattern
8. Number Triangle (1, 12, 123 ...)
9. Number Triangle (1, 22, 333 ...)
10. Floyd's Triangle
11. Pascal's Triangle
12. Number Pyramid (Centered)
13. Inverted Number Triangle
14. Alphabet Triangle (A, AB, ABC ...)
15. Alphabet Pyramid (Centered)
16. Hollow Square Pattern
17. Hollow Rectangle Pattern
18. Hollow Right Triangle Pattern
19. Hollow Pyramid Pattern
20. Hollow Diamond Pattern
21. Butterfly Pattern
22. Hourglass (Sandglass) Pattern
23. Zigzag Pattern
24. Reverse Number Pattern (Decreasing Rows)
25. Checkerboard (1/0) Pattern
26. Diamond Pattern with Numbers
27. Right-Aligned Decreasing Triangle
28. Concentric Square Number Pattern
29. Plus (+) Pattern
30. X (Cross) Pattern

2. Basic Programs

1. Sum of First N Natural Numbers

Output

```
Enter a number: 10
Sum = 55
```

2. Factorial of a Number

Output

```
Enter a number: 5
Factorial = 120
```

3. Multiplication Table

Output

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

4. Sum of Digits of a Number

Output

```
Enter a number: 12345
Sum of digits = 15
```

5. Reverse a Number

Output

```
Enter a number: 1234
Reversed number = 4321
```

6. Check Prime Number

Output

```
Enter a number: 29
29 is a Prime number
```

7. Fibonacci Series

Output

```
Enter number of terms: 10
0 1 1 2 3 5 8 13 21 34
```

8. Check Palindrome Number

Output

```
Enter a number: 121
121 is a Palindrome number
```

9. GCD of Two Numbers

Output

```
Enter two numbers: 36 60
GCD = 12
```

10. Print Even and Odd Numbers in a Range

Output

```
Even numbers:  
2 4 6 8 10 12 14 16 18 20  
Odd numbers:  
1 3 5 7 9 11 13 15 17 19
```

3. Pattern Programs

1. Right Triangle (Solid Star Pattern)

Output

```
*
* *
* * *
* * * *
* * * * *
```

2. Right-Aligned Triangle

Output

```

  *
 * *
* * *
* * * *
* * * * *

```

3. Inverted Right Triangle

Output

```
* * * * *
* * * *
* * *
* *
*
```

4. Inverted Right-Aligned Triangle

Output

```
* * * * *
 * * * *
  * * *
   * *
    *
```

5. Pyramid Pattern

Output

```

  *
 * *
* * *
* * * *
* * * * *
```

6. Inverted Pyramid Pattern

Output

```
* * * * *
 * * * *
  * * *
   * *
    *
```

7. Diamond Pattern

Output

```

  *
 * *
* * *
* * * *
* * * * *
* * * *
 * * *
  * *
   *
```

8. Number Triangle (1, 12, 123 ...)

Output

```
1
12
123
1234
12345
```

9. Number Triangle (1, 22, 333 ...)

Output

```
1
22
333
4444
55555
```

10. Floyd's Triangle

Output

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

11. Pascal's Triangle

Output

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

12. Number Pyramid (Centered)

Output

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

13. Inverted Number Triangle

Output

```
5 4 3 2 1
4 3 2 1
3 2 1
2 1
1
```

14. Alphabet Triangle (A, AB, ABC ...)

Output

```
A
AB
ABC
ABCD
ABCDE
```

15. Alphabet Pyramid (Centered)

Output

```
  A
 A B
A B C
A B C D
A B C D E
```

16. Hollow Square Pattern

Output

```
* * * * *
*           *
*         *
*       *
*     *
* * * * *
```

17. Hollow Rectangle Pattern

Output

```
*****
*       *
*       *
*       *
*****
```

18. Hollow Right Triangle Pattern

Output

```
*
**
* *
* *
****
```

19. Hollow Pyramid Pattern

Output

```
  *
 * *
*   *
* * *
*****
```

20. Hollow Diamond Pattern

Output

```
  *
 * *
*   *
* * *
* * *
*   *
 * *
  *
```

21. Butterfly Pattern

Output

```
*           *
**        **
***      ***
****    ****
*****
****    ****
***      ***
**        **
*           *
```

22. Hourglass (Sandglass) Pattern

Output

```
 * * * * *
  * * * *
   * * *
    * *
     *
    * *
   * * *
  * * * *
 * * * * *
```

23. Zigzag Pattern

Output

```
 *   *   *
 * * * * *
  *   *   *
```

24. Reverse Number Pattern (Decreasing Rows)

Output

```
5 4 3 2 1
4 3 2 1
3 2 1
2 1
1
```

25. Checkerboard (1/0) Pattern

Output

```
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
```

26. Diamond Pattern with Numbers

Output

```
  1
 1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4
 1 2 3
  1 2
   1
```

27. Right-Aligned Decreasing Triangle

Output

```
 * * * * *
  * * * *
   * * *
    * *
     *
    * *
```

28. Concentric Square Number Pattern

Output

```
5 5 5 5 5
5 4 4 4 5
5 4 3 4 5
5 4 4 4 5
5 5 5 5 5
```


29. Plus (+) Pattern

Output

```
  *
  *
*****
  *
  *
```

30. X (Cross) Pattern

Output

```
*   *
* *
 *
 *
* *
*   *
```